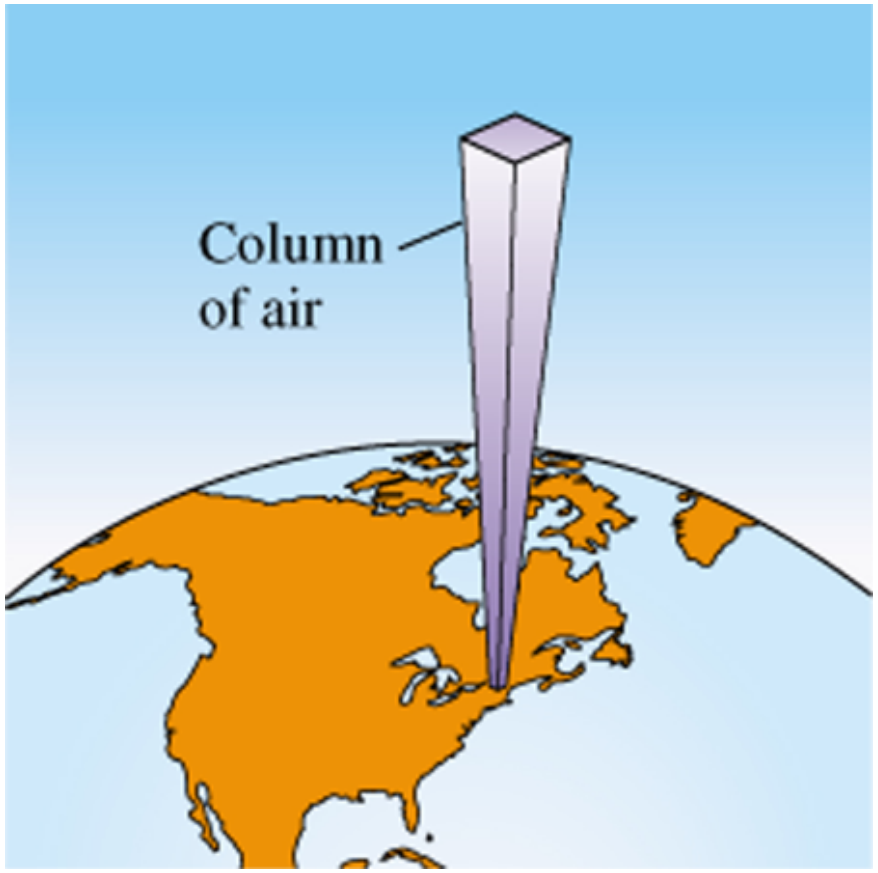
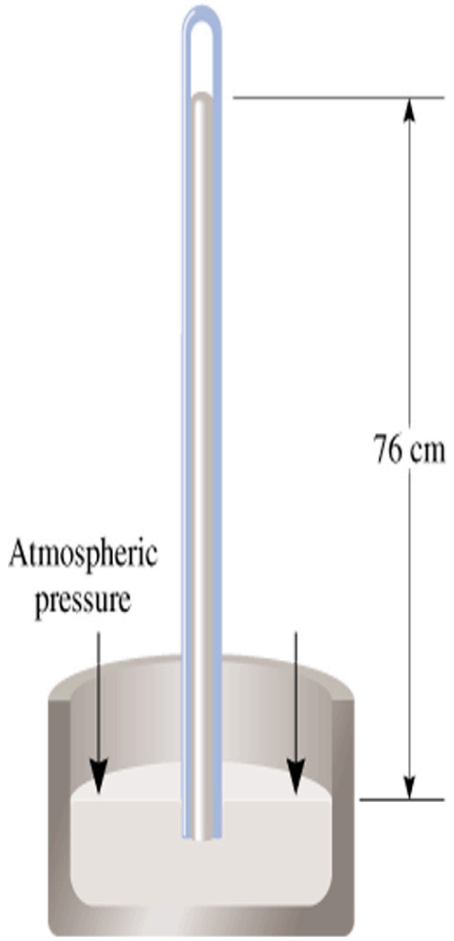
Pressure measurement

Pressure = Force / Area

- **Absolute pressure** is zero-referenced against a perfect vacuum, so it is equal to gauge pressure plus atmospheric pressure.
- **Gauge pressure** is zero-referenced against ambient air pressure, so it is equal to absolute pressure minus atmospheric pressure.
- **Differential pressure** is the difference in pressure between two points

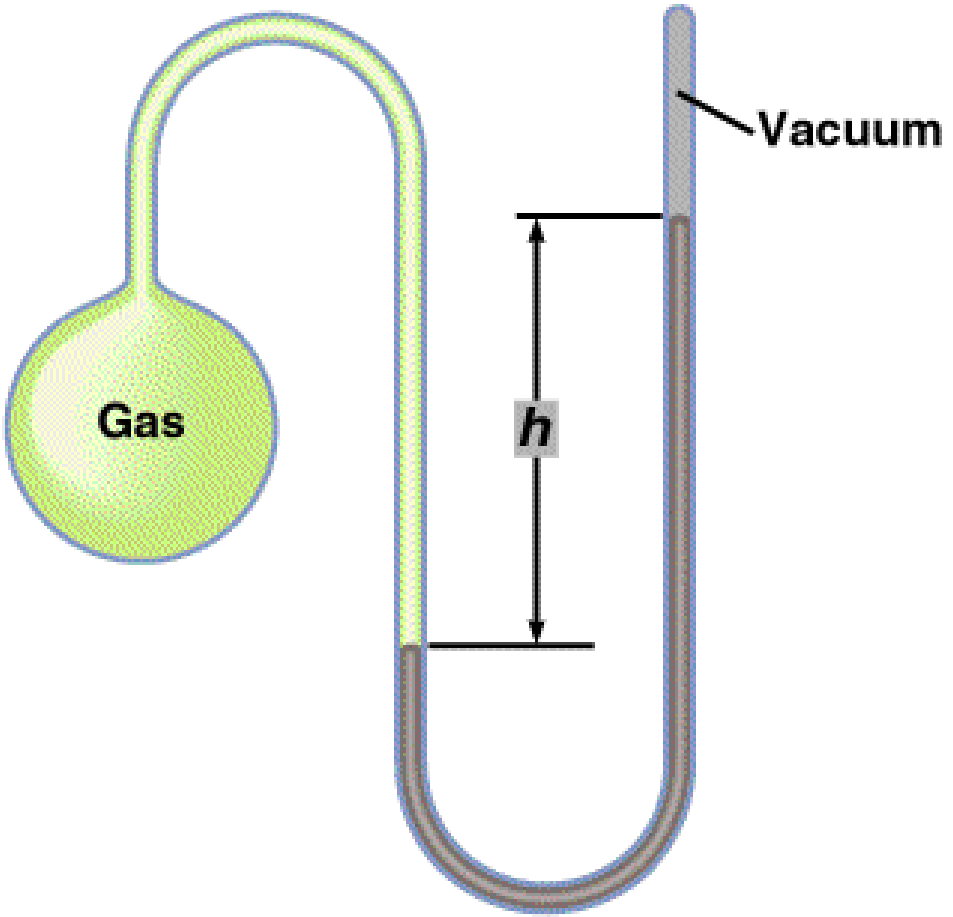




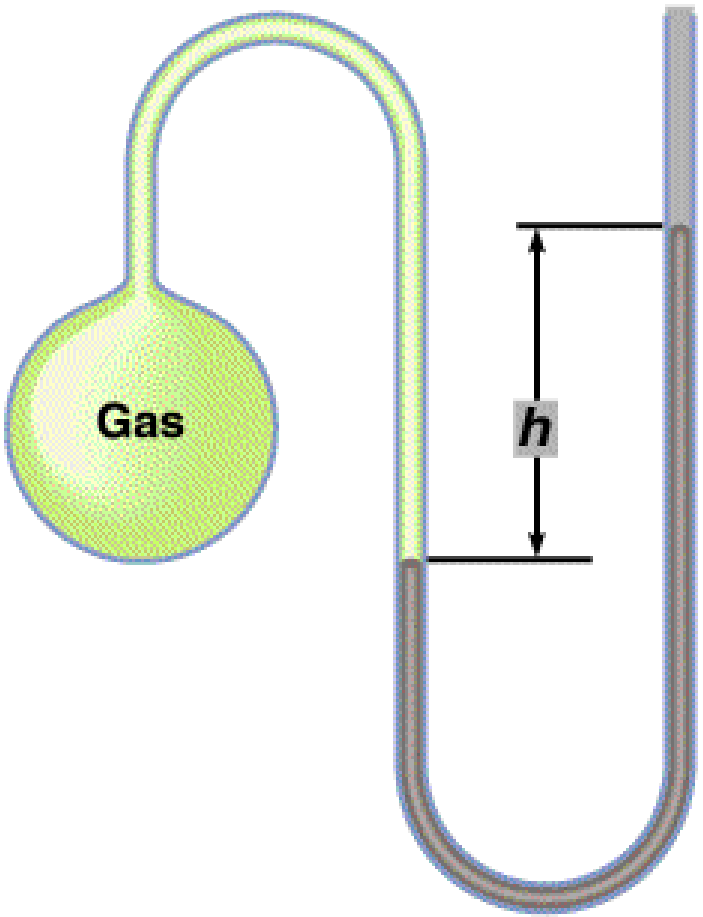
Barometer

faculty.uncfsu.edu/dautrey/CHEM%20140/chapter10-notes.ppt

Manometers Used to Measure Gas Pressures



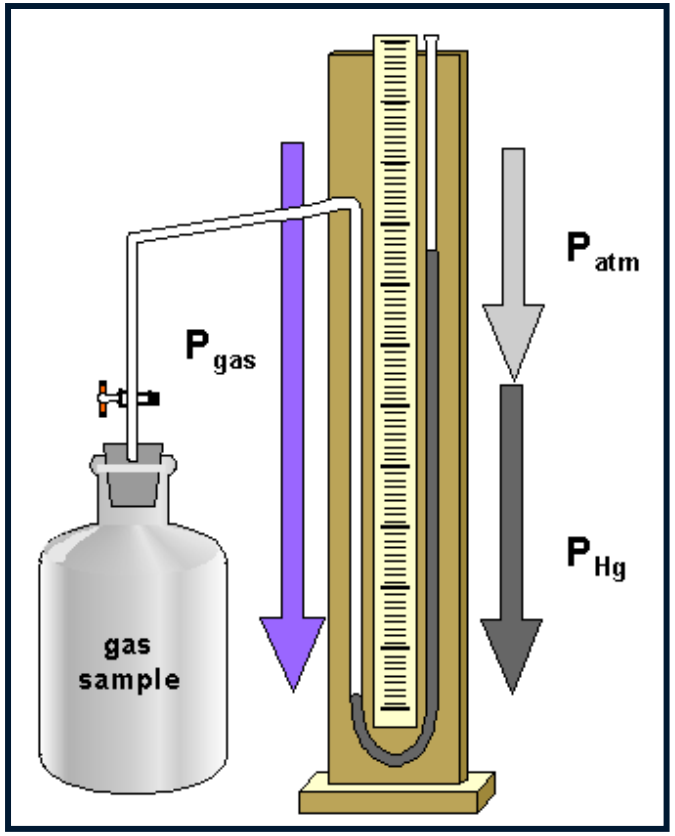
$P_{gas} = P_h$



$P_{gas} = P_h + P_{atm}$

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Manometer



U-tube Manometer

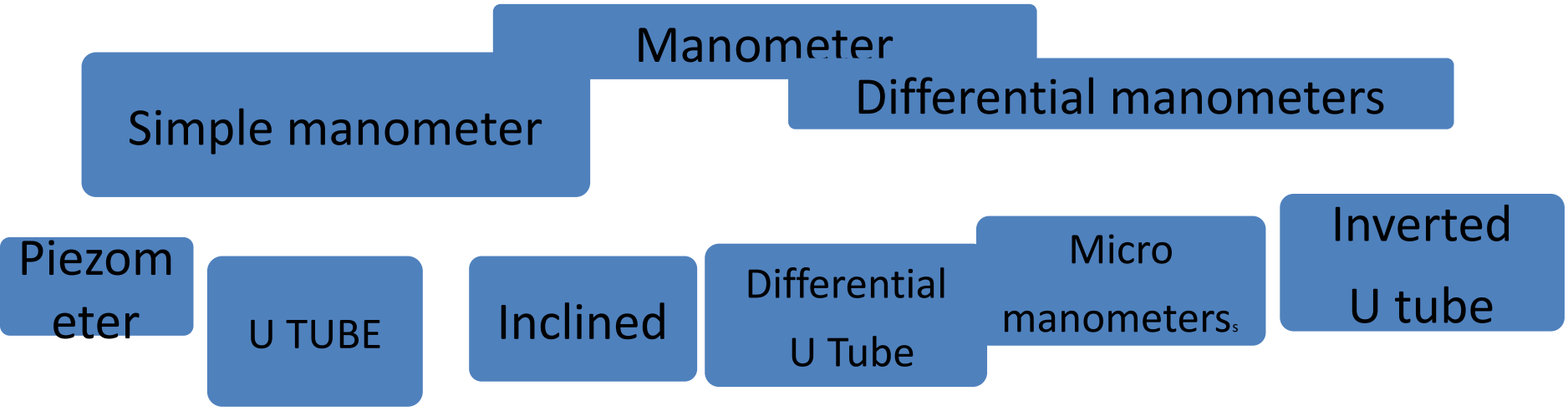
Courtesy Christy Johannesson www.nisd.net/communicationsarts/pages/chem

www.unit5.org/christjs/Gas%20Laws/Manometer.ppt

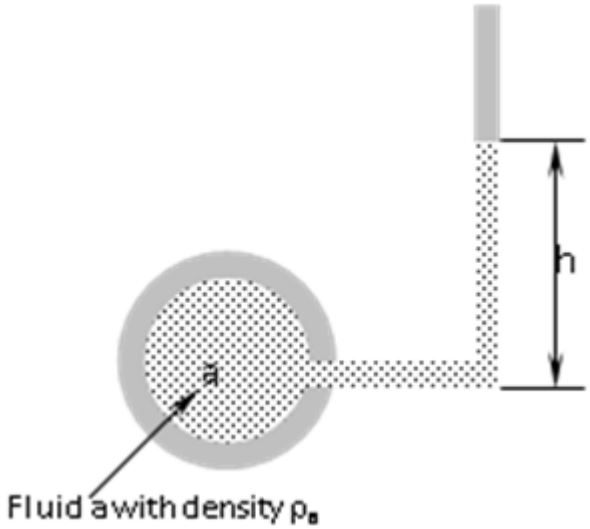
Manometer

www.unit5.org/christjs/Gas%20Laws/Manometer.ppt

Manometers are the devices used for measuring of pressure at a point in a fluid by balancing the column of fluid by the same or another column of fluid. They are classified as

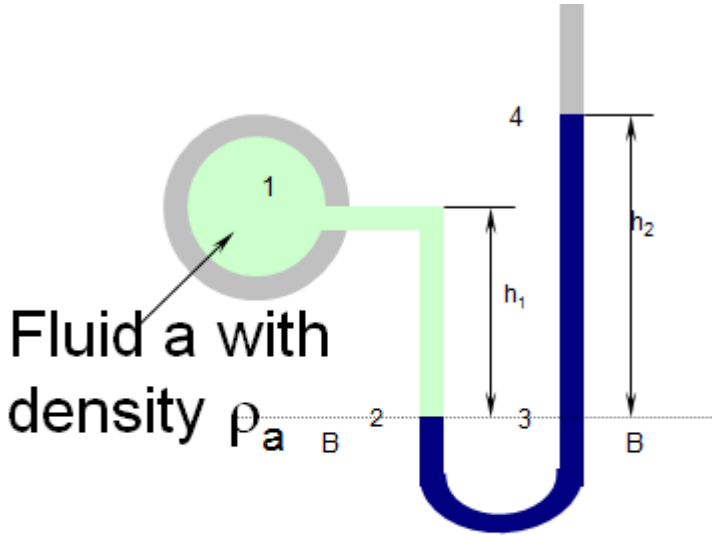


Piezometer: This is the simplest form of manometer used for measuring gauge pressures. One end of this is connected to the point where pressure is to be measured and other end is open to the atmosphere as shown in Figure. The rise of liquid, gives the pressure head at that point. If at a point A, the height of liquid say water is 'h' in piezometer tube, then the pressure at A



$P_a = \rho_a \cdot g \cdot h$

Simple U Tube Manometer



Simple 'U' tube Manometer to measure pressure at a point.

Starting with point 1.

$1 \rightarrow P_a$

$2 \rightarrow P_a + \rho_a g h_1$

The pressure at point 2 and 3 are same because the datum line is parallel to the earth's surface.

$\therefore 3 \rightarrow P_a + \rho_a g h_1$

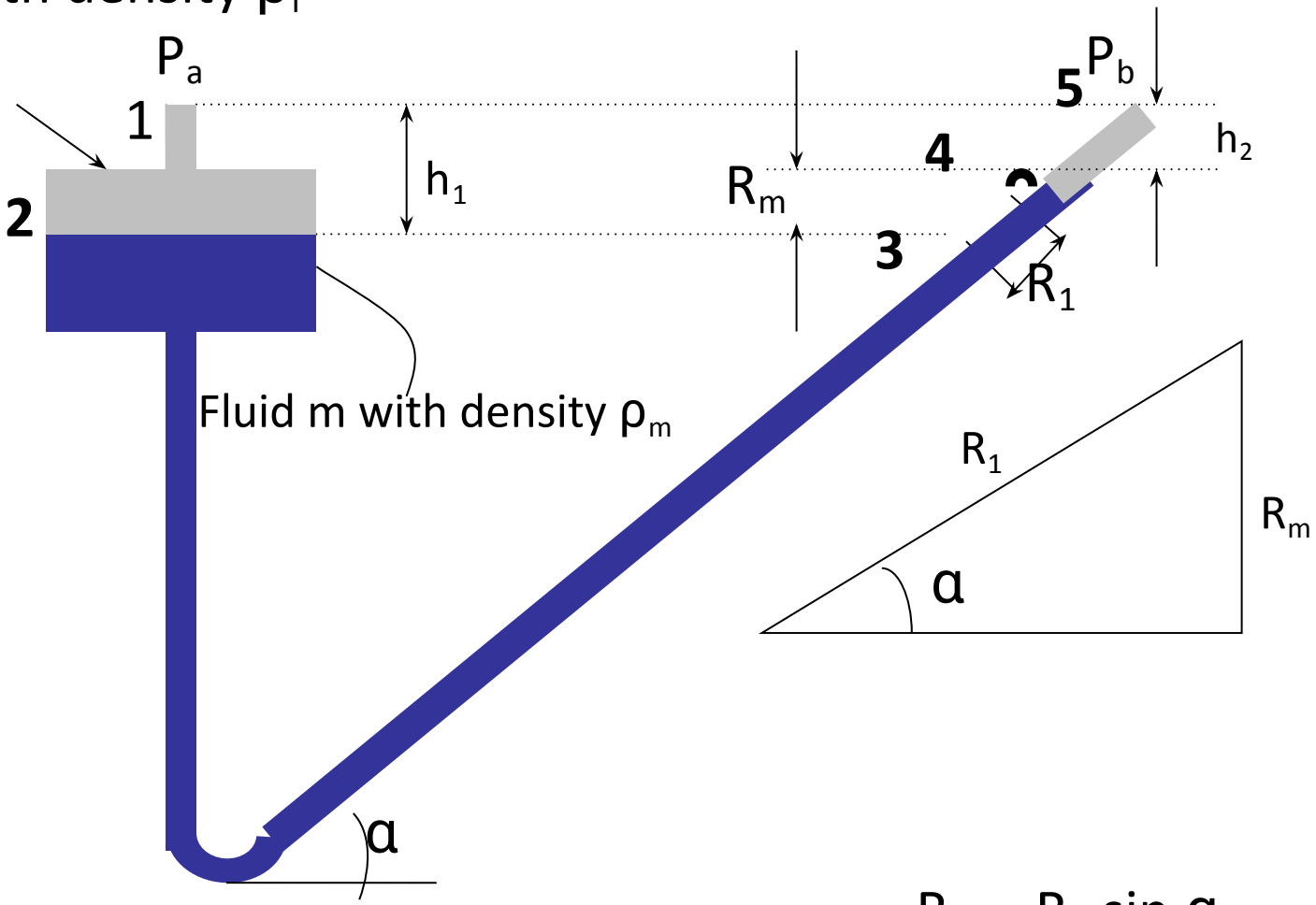
$4 \rightarrow P_a + \rho_a g h_1 - \rho_b g h_2 = 0$, because point 4 is exposed to atmosphere.

Now Pa will be

$Pa = \rho b\, g\, h_2 - \rho a\, g\, h_1$

Inclined manometer:

Fluid of with density ρ_f



Inclined manometer

$R_m = R_1 \sin \alpha$

Inclined manometer: In this type, one leg of the manometer is inclined in such a manner that for a small magnitude of R_m the meniscus in the inclined tube will move considerable distance along the tube. This distance R_1, is R_m divided by the sine of α, the angle of inclination.

By making 'α' small, the magnitude of R_m is multiplied into a long distance R_1 and a large reading becomes equivalent to a small pressure difference.

To derive relationship between ($P_a - P_b$) and R_m, we start from point 1

$1 \rightarrow P_a$

$2 \rightarrow P_a + \rho_f g h_1$

Pressure at point 3 is equal to pressure at point 2, from hydrostatic principles

$3 \rightarrow P_a + \rho_f g h_1$

$4 \rightarrow P_a + \rho_f g h_1 - \rho_m g R_m$

$5 \rightarrow P_a + \rho_f g h_1 - \rho_m g R_m - \rho_f g h_2 = P_b$

By simplifying the above equation

$P_a - P_b = \rho_m g R_m - \rho_f g h_1 + \rho_f g h_2$

$P_a - P_b = \rho_m g R_m + \rho_f g(h_2 - h_1)$

$P_a - P_b = \rho_m g R_m + \rho_f g(-R_m)$

$P_a - P_b = g R_m (\rho_m - \rho_f)$

From the figure

$R_m = R_1 \sin \alpha$

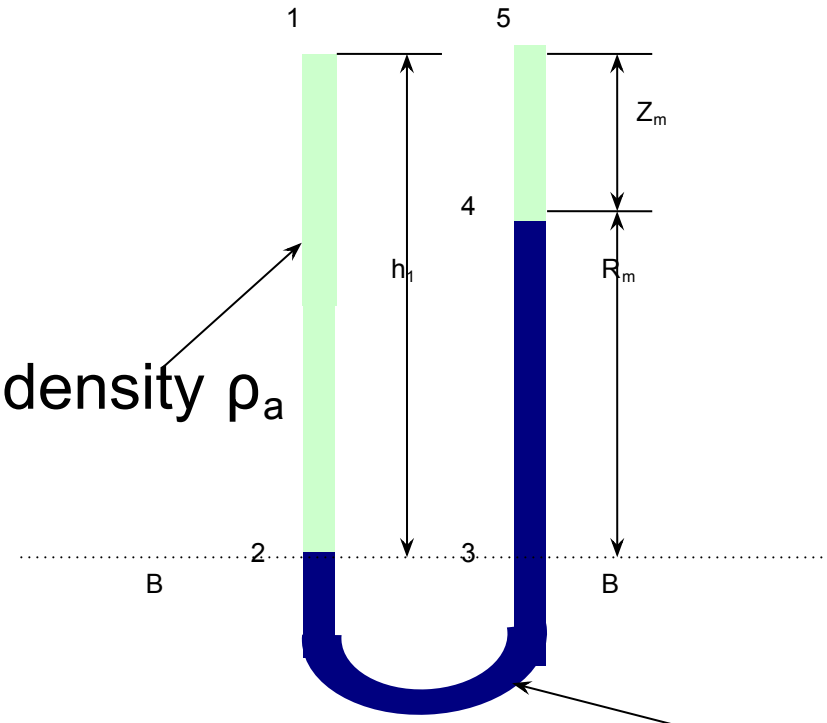
Substituting the value of R_m in the above equation gives

$$P_a - P_b = g R_1 \sin \alpha (\rho_m - \rho_f)$$

In this type of pressure gauge, it is necessary to provide an enlargement in the vertical leg so that the variation in the level of fluid in the enlargement is negligible

U’ tube manometer *[to measure pressure difference between two points]*:

A ‘U’ tube manometer is shown in figure . It consists of a trasparent U-tube containing the fluid A whose pressure is to be measured and an immiscible fluid B of higher density. The limbs are connected to two points between which the pressure difference is required. The connecting leads should be completely full of fluid A.



Fluid a with density ρ_a

Manometric fluid b with density ρ_b

Simple ‘U’ tube Manometer to measure pressure difference

The Pressure Pa is excerted in one arm of the U- tube and a pressure P_b in the other. As a result of the difference in pressure $P_a - P_b$, the meniscus in one branch of the U- tube is higher than in the other, and the vertical distance between the two meniscus R_m may be to measure the difference in pressure.

To derive the relationship between P_a - P_b and R_m, we start with point 1.

1 $\rightarrow P_a$

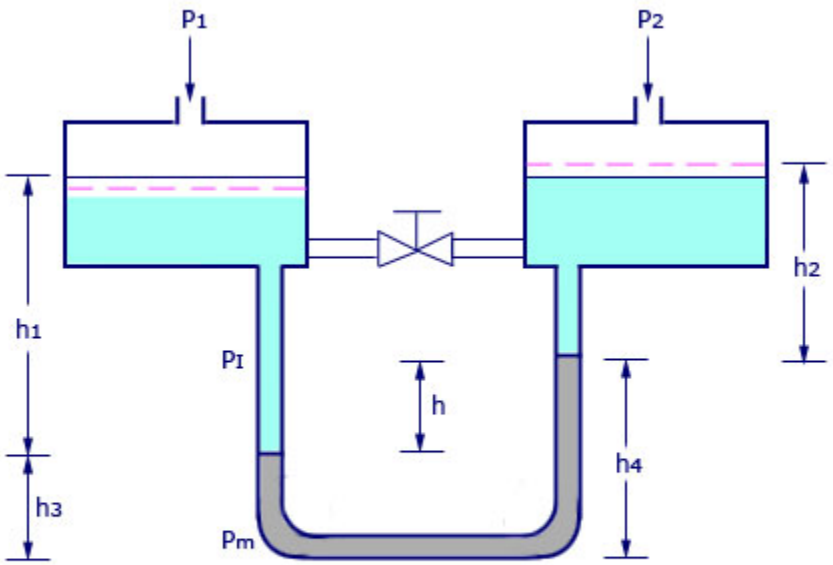
2 $\rightarrow P_a + \rho_a g h_1 = P_a + \rho_a g (Z_m + R_m)$ { since h1 = Zm + Rm, refer figure }

by principles of hydrostatics, pressure at point 2 = pressure at point 3

$$3 \rightarrow P_a + \rho_a g (Z_m + R_m)$$
$$4 \rightarrow P_a + \rho_a g (Z_m + R_m) - \rho_b g R_m$$
$$5 \rightarrow P_a + \rho_a g (Z_m + R_m) - \rho_b g R_m - \rho_a g Z_m = P_b$$
$$\Rightarrow P_a - P_b = - \rho_a g (Z_m + R_m) + \rho_b g R_m + \rho_a g Z_m$$
$$\Rightarrow P_a - P_b = g R_m (\rho_b - \rho_a)$$

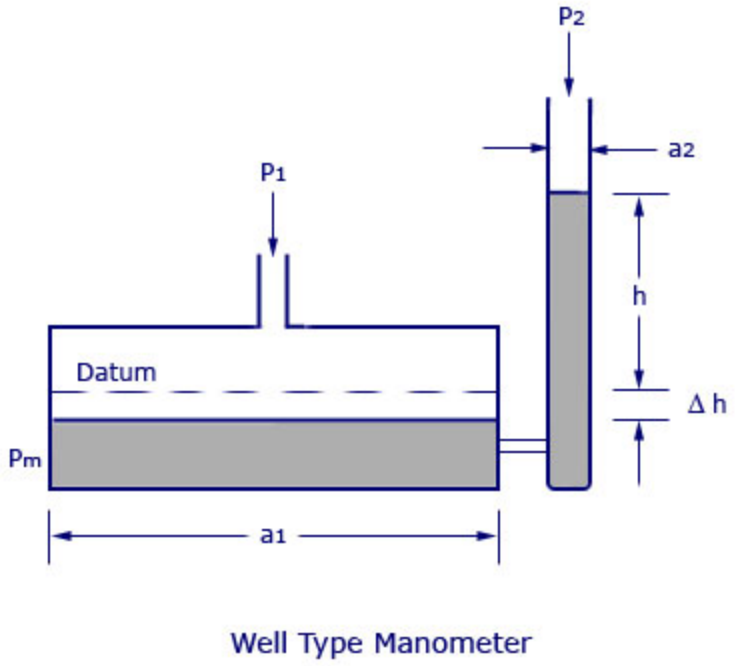
This relationship is independent of the distances Z_m and the dimensions of the tube provided that pressure P_a and P_b are measured in the same horizontal plane

Micro manometer, two fluid manometer , Well type of manometer



Manometer With Large Seal Pots

www.InstrumentationToday.com

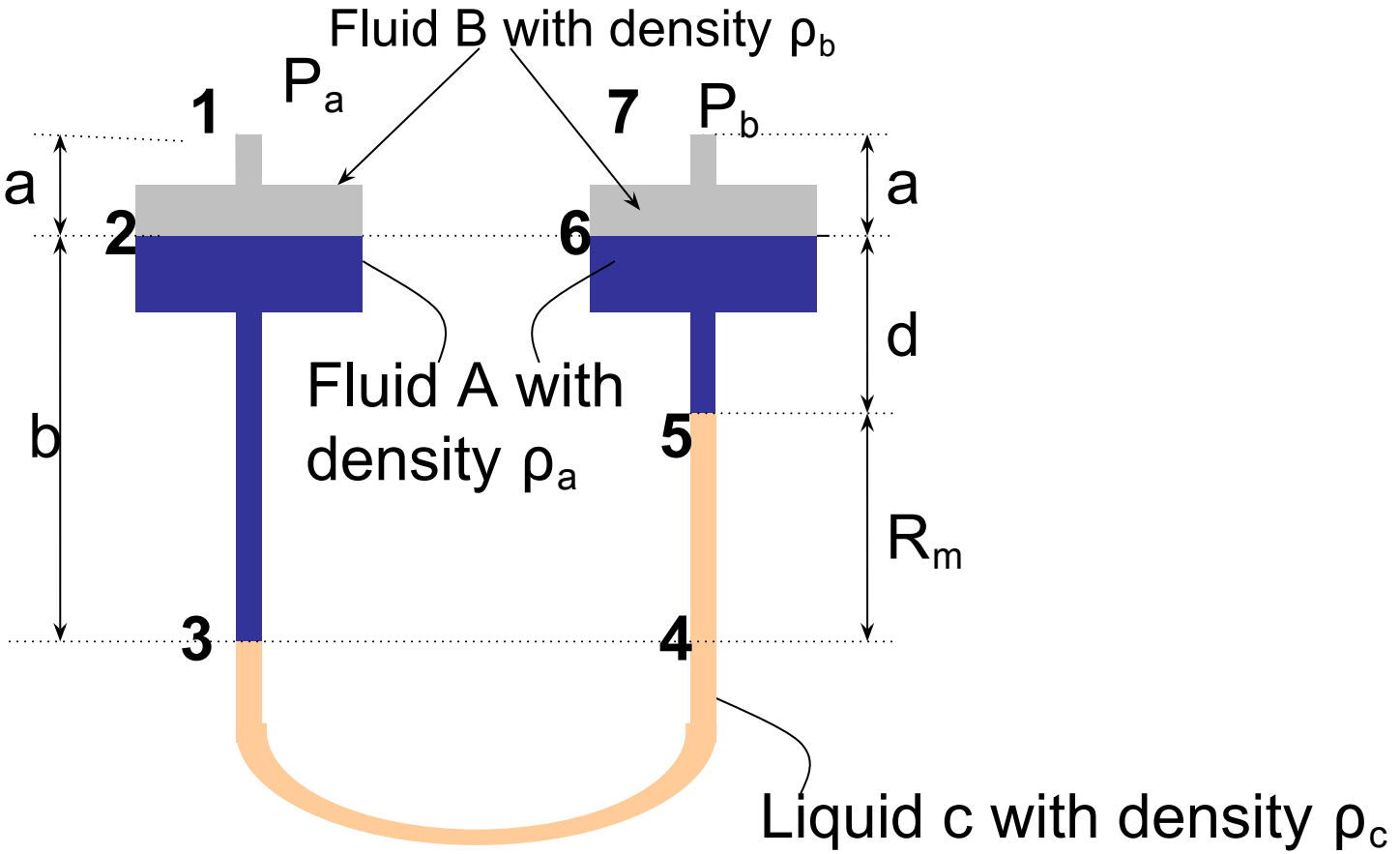


Well Type Manometer

www.InstrumentationToday.com

Differential manometer [Micro manometer, two fluid manometer , Well type of manometer]

For the measurement of smaller pressure differences, this type of manometer as shown in the figure is often used.



Two fluid Micro manometer

- This manometer contains two liquids a and c which are immiscible and whose densities vary as small as as possible.
- Enlarged chambers are inserted in the manometer, so that the level of fluid at point 2 and 6 does not change appreciably with changes in the reading R_m. Hence the distance between points 1 and 2 may be considered equal to the distance between points 6 and 7.
- The hydrostatic principle may be employed in developing the relationship between the reading of this type manometer and the pressure difference $P_a - P_b$
- The changes in pressure from the points 1 to 7 are :

1 $\rightarrow P_a$

2 $\rightarrow P_a + \rho_b g a$

3 $\rightarrow P_a + \rho_b g a + \rho_a g b$

Pressure at point 3 is equal to pressure at point 4, from hydrostatic principles

4 $\rightarrow P_a + \rho_b g a + \rho_a g b$

5 $\rightarrow P_a + \rho_b g a + \rho_a g b - \rho_c g R_m$

6 $\rightarrow P_a + \rho_b g a + \rho_a g b - \rho_c g R_m - \rho_a g d$

7 $\rightarrow P_a + \rho_b g a + \rho_a g b - \rho_c g R_m - \rho_a g d - \rho_b g a = P_b$

Simplifying the above equation we get,

$P_a - P_b = \rho_a g (d - b) + \rho_c g R_m$

From figure, b-d = Rm

$\therefore P_a - P_b = R_m g (\rho_c - \rho_a)$

From this equation it follows that the smaller the difference $(\rho_c - \rho_a)$, larger will be the reading of R_m on the manometer for the given value of ΔP.

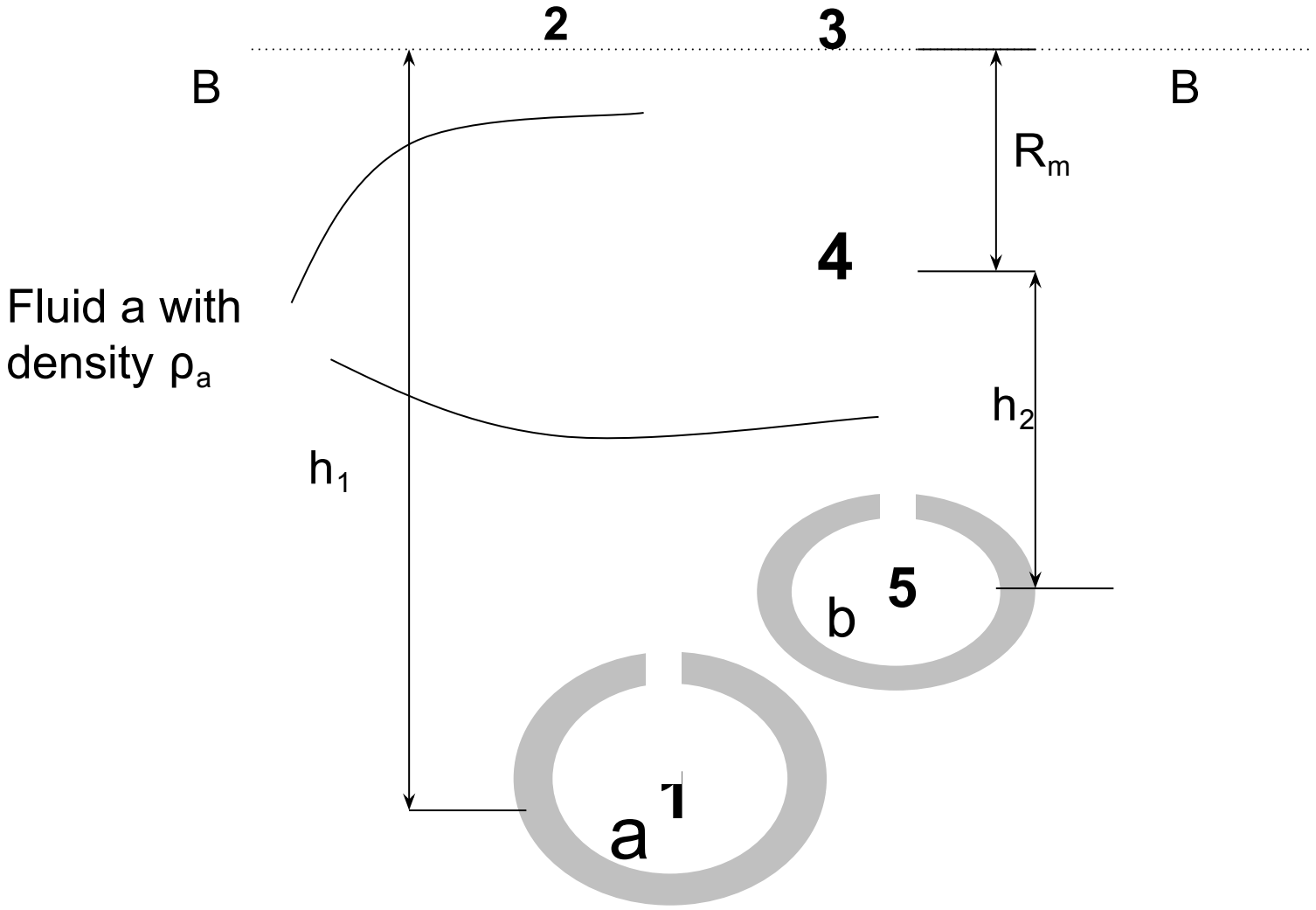
Inverted manometer:(Applied when the density of the manometric fluid is less than the fluid in the measuring points)

It consists of an inverted 'U' tube containing a light liquid. The two ends of the tube are connected to the points whose difference of pressure is to be measured.

It is used for measuring low-pressure differences. An inverted U tube manometer connected to two points A and B is shown in Figure

Let the pressure at A is more than the pressure at B. Let h_1 be the height of liquid in left limb below the datum line B-B and h_2 be the height of liquid in right limb.

R_m be the manometric reading. Let ρ_a and ρ_b be the density of liquid A and manometric liquid B respectively.



Inverted 'U' tube Manometer to measure low pressure differences

Starting from point 1:

$1 \rightarrow P_a$

$2 \rightarrow P_a - \rho_a g h_1$

Pressure at point 3 is equal to pressure at point 2, from hydrostatic principles

$3 \rightarrow P_a - \rho_a g h_1$

$4 \rightarrow P_a - \rho_a g h_1 + \rho_b g R_m$

$5 \rightarrow P_a - \rho_a g h_1 + \rho_b g R_m + \rho_a g h_2 = P_b$

Solving for $P_a - P_b$

$P_a - P_b = \rho_a g h_1 - \rho_b g R_m - \rho_a g h_2$

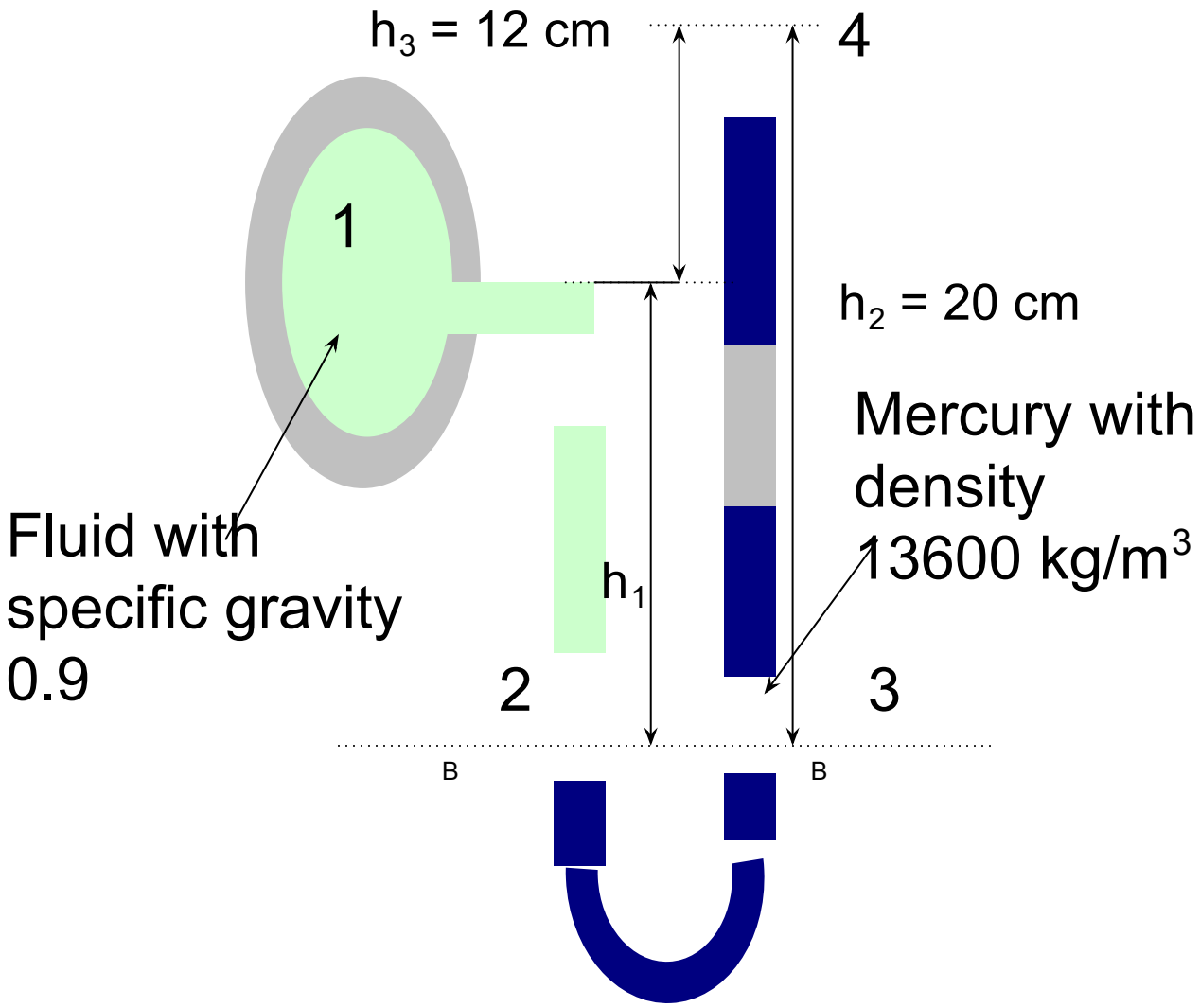
Substituting the value of h_1 in the above equation, we have

$P_a - P_b = \rho_a g (R_m + h_2) - \rho_b g R_m - \rho_a g h_2$

$P_a - P_b = \rho_a g (h_1 – h_2) –\rho_b g R_m$

Numerical Problems

1. The right limb of a simple U tube manometer containing mercury is open to the atmosphere, while the left limb is connected to a pipe in which a fluid of specific gravity 0.9 is flowing. The centre of the pipe is 12 cm below the level of mercury in the right limb. Find the pressure of the fluid in the pipe if the difference of mercury level in two limbs is 20cm.


h3 = 12 cm
4
1
h2 = 20 cm
Mercury with
density
13600 kg/m3
Fluid with
specific gravity
0.9
h1
2
3
B
B

The data are represented in the figure.

From the figure

$h_1 = h_2 – h_3$

$h_1 = 20 – 12 = 8$ cm

$h_1 = 0.08$m.

The pressure at different points are written as

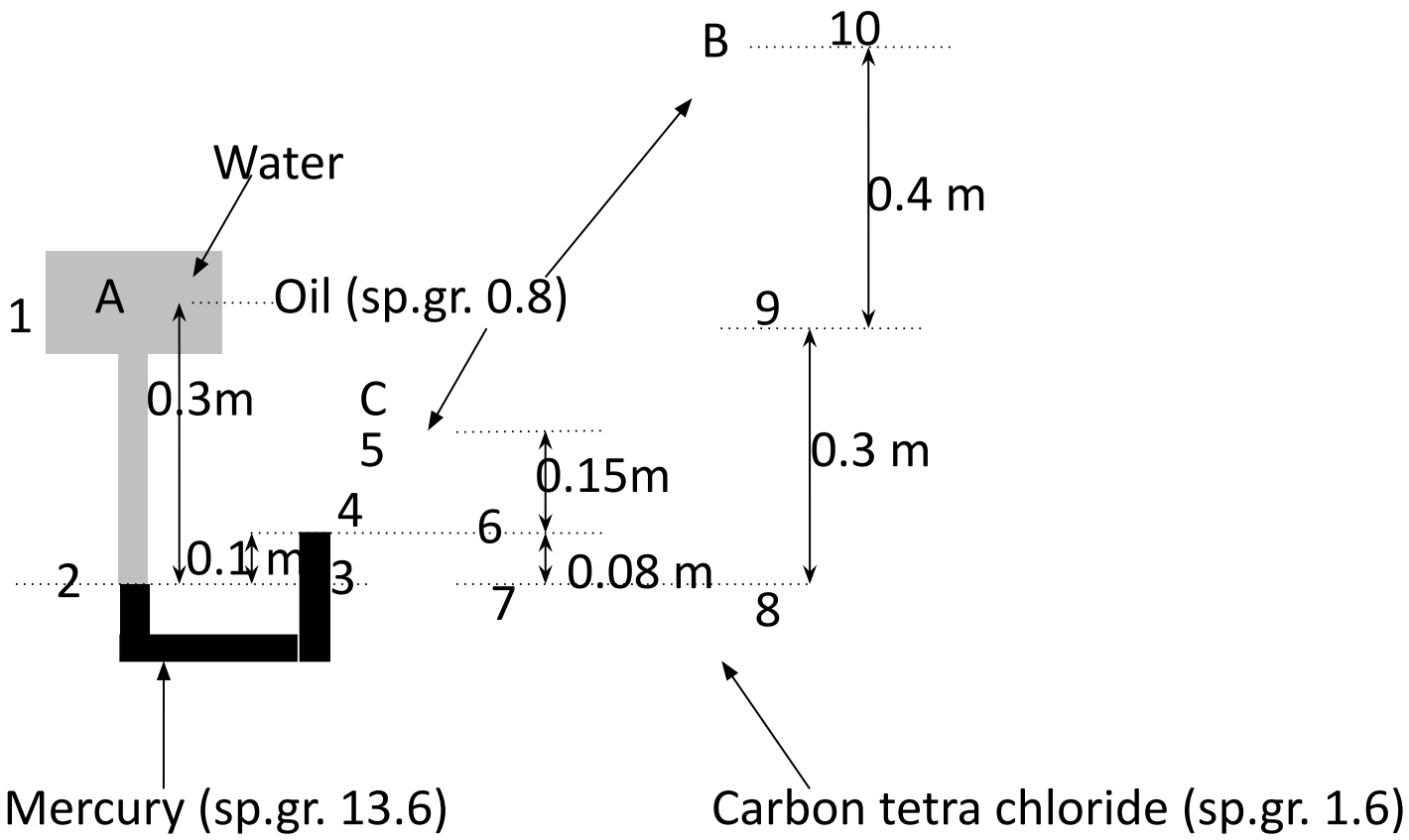
1. P_1
2. $P_1 + \rho_a g h_1$
3. $P_1 + \rho_a g h_1$
4. $P_1 + \rho_a g h_1 - \rho_m g h_2 = 0$

$P_1 = g[\rho_m h_2 - \rho_a h_1]$

$= 9.81*[13600*0.20 – 900*0.08]$

$P_1 = 25976.88\ \mathrm{N/m^2}$

Determine ($P_A - P_B$) and ($P_A - P_c$) in the figure



The pressure at different points are written to find the difference between (P_A-P_C) as

1. P_A
2. $P_A + \rho_w * g * 0.3$
3. $P_A + \rho_w * g * 0.3$
4. $P_A + \rho_w * g * 0.3 - \rho_m * g * 0.1$
5. $P_A + \rho_w * g * 0.3 - \rho_m * g * 0.1 - \rho_o * g * 0.15 = P_c$

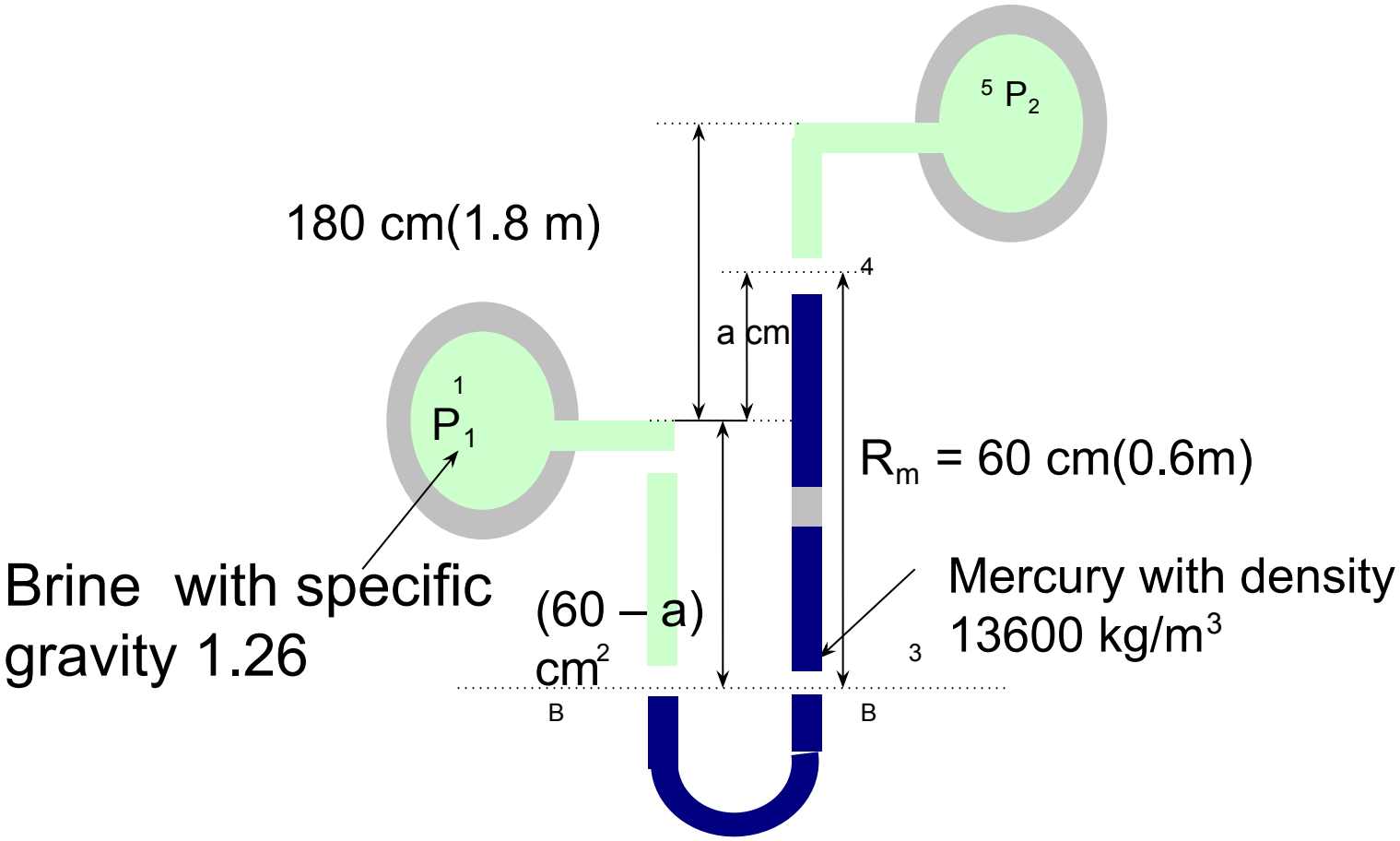
$P_A - P_C = \rho_m * g * 0.1 + \rho_o * g * 0.15 - \rho_w * g * 0.3$

$P_A - P_C = 13600 * 9.81 * 0.1 + 800 * 9.81 * 0.15 - 1000 * 9.81 * 0.3$

$= 11500 \text{N/m}^2$

$P_A - P_B$ = 17610 N/m^2

1. Two tubes running full with brine(sp. gr. 1.26) are connected to a differential mercury manometer. The connecting tubes are filled with water. The centre line of high pressure pipe is 180 cm lower in elevation than the centre line of the other pipe. If the mercury reading is 60 cm, what is the pressure difference in N/m^2.


5 P2
180 cm(1.8 m)
4
a cm
1
P1
Rm = 60 cm(0.6m)
Brine with specific gravity 1.26
(60 – a) cm2
Mercury with density 13600 kg/m3
3
B
B

Let the distance between mercury level and the lower tube be (points 1- 4) a.

$\therefore$ the distance between points 1 and 2 is (0.6-a).

The pressure at different points are written as

1. P_1
2. $P_1 + \rho_b * g * (0.60-a)$
3. $P_1 + \rho_b * g * (0.60-a)$
4. $P_1 + \rho_b * g * (0.60-a) - \rho_m * g * 0.60$
5. $P_1 + \rho_b * g * (0.60-a) - \rho_m * g * 0.60 - \rho_b * g * (1.80-a) = P_5$

$\therefore P_1 - P_5 = \rho_m * g * 0.60 + \rho_b * g * (1.80-a) - \rho_b * g * (0.60-a)$

$P_1 - P_5 = \rho_m * g * 0.60 + \rho_b * g * 1.80 - \rho_b * g * a - \rho_b * g * 0.60 + \rho_b * g * a$

Solving

$P_1 - P_5 = \rho_m * g * 0.60 + \rho_b * g * 1.80 - \rho_b * g * 0.60$

$P_1 - P_5 = 13600 * 9.81 * 0.60 + 1000 * 9.81 * 1.80 - 1000 * 9.81 * 0.60$

$\mathbf{P_1 - P_5 = 91728\ N/m^2.}$

Exercise problem

1.An open tank contains water upto a depth of 1.5m and above it an oil of specific gravity 0.8 for a depth of 2m. Find the pressure intensity ;

at the interface of the two liquids

at the bottom of the tank

2. U tube with both ends open to atmosphere contains mercury in the lower portion. In one limb water stands to a height of 107.6 cm above the surface of mercury .In the other limb oil of specific gravity 0.8 stands to a height of 80.5 cmabove the surface of mercury.Find the difference in the levels of the top surface of water and oil.